

Exploring new directions in enhancing the ACTS parameter optimization suite

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Track seeding strongly affects both the quality and computational cost of charged-particle reconstruction, yet its many configuration parameters are commonly tuned through expert intuition and repeated trial and error. ACTS reduces this burden with an Optuna Tree-structured Parzen Estimator auto-tuner, but expensive evaluations, a restricted search space, and a scalarized objective can limit evaluation efficiency, exclude promising configurations, and obscure performance trade-offs. We investigate whether Bayesian optimization can address these limitations using ACTS with the Open Data Detector (ODD). Under identical search ranges and a common 100-trial budget, we compare Expected Improvement and Upper Confidence Bound with TPE and random search on the existing eight-parameter problem, extend the best-performing Bayesian method to fifteen parameters, and apply Expected Hypervolume Improvement to optimize efficiency, fake rate, duplicate rate, and runtime without fixed scalar weights. Candidate configurations are evaluated through the full ACTS reconstruction chain and validated on disjoint held-out events. The Bayesian acquisition methods identify strong configurations earlier than TPE, and their advantage persists in held-out validation. Expanding the search further improves performance, while multi-objective optimization reveals competitive non-dominated solutions spanning distinct trade-offs. These results indicate that Bayesian optimization can strengthen ACTS auto-tuning through efficient evaluation, broader parameter searches, and post-hoc expert selection among non-dominated alternatives.

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1. Introduction

Track seeding strongly affects both charged-particle reconstruction quality and downstream computing cost. Seeding algorithm parameters are commonly tuned through expert intuition and repeated trial and error, a time-consuming process that depends on detector geometry and operating conditions [1].

ACTS [2] reduces this burden with an Optuna [3] Tree-structured Parzen Estimator (TPE) [4] auto-tuner that maximizes a scalarized score over eight seeding parameters. Yet three gaps remain: full-chain trials make evaluation efficiency essential; the eight-parameter space may exclude better configurations; and fixed scalarization weights hide viable performance trade-offs.

We test Bayesian optimization against these gaps. Expected Improvement (EI) and Upper Confidence Bound (UCB) are compared with TPE and random search on the same eight-parameter problem, then EI is extended to fifteen parameters. Expected Hypervolume Improvement (EHVI) optimizes the metrics separately and returns a Pareto set for post-hoc selection.

2. Experimental setup

2.1 ACTS, ODD, and evaluation

We use A Common Tracking Software (ACTS), an open-source reconstruction framework, with the Open Data Detector (ODD), a realistic public HL-LHC testbed [5]. We simulate $t\bar{t}$ events at $\sqrt{s} = 14$ TeV with 200 pile-up interactions. Each event, represented as a candidate vector $\mathbf{x}$, is processed through the full ACTS chain on a fixed ten-event training sample, over which we average efficiency $\epsilon(\mathbf{x})$, fake rate $f(\mathbf{x})$, duplicate rate $d(\mathbf{x})$, and runtime $t(\mathbf{x})$. Selected configurations are evaluated on validation events to check overfitting. Figures report training behavior; tables report performance on validation events.

2.2 Optimization methods

All methods use the same sequential propose–evaluate–update loop. At iteration n , the optimizer selects a seeding vector $\mathbf{x}_n$; ACTS returns (ϵ, f, d, t) or its scalarized score, and the observation is appended to the trial history $\mathcal{D}_n$. TPE refits densities to better and worse observations [4], whereas Bayesian optimization refits a probabilistic surrogate and scores candidates with an acquisition function. It then selects $\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} a(\mathbf{x})$. Thus, both the surrogate and acquisition optimum are updated after every ACTS evaluation. Xopt v2.4 implements this Bayesian loop [6], which repeats until the trial budget is exhausted.

Single-objective optimization: Single-objective methods require the metrics to be scalarized. We use the ACTS auto-tuning score [7, 8], so all methods maximize the same objective:

$$S(\mathbf{x}) = \epsilon(\mathbf{x}) - [f(\mathbf{x}) + d(\mathbf{x})/5 + t(\mathbf{x})/5]. \quad (1)$$

The Bayesian surrogate provides a predicted score $\mu(\mathbf{x})$ and uncertainty $\sigma(\mathbf{x})$. Expected Improvement (EI) [9] estimates the gain over the best observed score S^* ; Upper Confidence Bound (UCB) [10] rewards predicted score and uncertainty:

$$a_{\text{EI}}(\mathbf{x}) = (\mu - S^*)\Phi(z) + \sigma\phi(z), \quad z = (\mu - S^*)/\sigma, \quad (2)$$

$$a_{\text{UCB}}(\mathbf{x}) = \mu(\mathbf{x}) + \beta\sigma(\mathbf{x}). \quad (3)$$

Here Φ and ϕ are the standard-normal cumulative distribution and density; β controls exploration, and we use $\beta = 2$ in this work.

Multi-objective optimization: Multi-objective Bayesian optimization treats efficiency, fake rate, duplicate rate, and runtime as separate objectives rather than combining them into the scalarized score in Eq. (1). Expected Hypervolume Improvement (EHVI) [11] favors candidate configurations predicted to improve the current set of best trade-off solutions. This allows the operating point to be selected after comparing the resulting trade-offs.

2.3 Experiments

Each experiment targets one Section 1 limitation using the Section 2.1 evaluation chain. Within each comparison, all optimizers use the same initialization, bounds, and event samples.

Evaluation efficiency: The first experiment compares how efficiently different optimizers use a fixed trial budget to tune ACTS seeding parameters. The eight parameters are `maxSeedsPerSpM`, `deltaRMin`, `deltaRMax`, `impactMax`, `sigmaScattering`, `radLengthPerSeed`, `cotThetaMax`, `maxPtScattering`. We run TPE, EI, UCB, and a uniform-random baseline for 100 single-objective trials. Running-best score measures convergence; each method’s best configuration is then tested on validation events.

Search-space scaling: To address limited tuning scope, the second experiment extends the problem from eight to fifteen seeding parameters using EI, adding `rMin`, `rMax`, `zMin`, `zMax`, `collisionZMin`, `collisionZMax`, `minPt`. Across 200 single-objective trials, we test whether the expanded set yields higher scores.

Optimization without scalarization: The third experiment explores tuning without scalarization. Over 250 trials, EHVI treats the four metrics as separate objectives while tuning the original eight parameters. Representative Pareto-front configurations are validated for post-hoc expert selection.

3. Results

3.1 Evaluation efficiency on the eight-parameter problem

Within the common 100-trial budget, EI reaches its best observed training score before UCB and TPE, while UCB reaches its best before TPE (Figure 1). Validation scores preserve this ordering: 92.29 for EI, 92.13 for UCB, and 91.99 for TPE (Table 1). Earlier convergence to higher-scoring configurations shows that the Bayesian acquisition methods use the limited evaluation budget more efficiently than TPE. All three optimizers outperform the random baseline.

3.2 Scaling to fifteen parameters

The fifteen-parameter EI optimization achieves a validation single-objective score of 92.73, an improvement of 0.44 over the best eight-parameter result (Table 1). The selected configuration increases efficiency by 0.43 percentage points and reduces runtime by 48.7%, with the duplicate rate rising from 15.51% to 16.83%. The higher score indicates that the additional geometric and kinematic controls expose configurations beyond those accessible to the original search.

3.3 Optimization without scalarization

Without fixing metric weights in advance, EHVI identifies a Pareto front of attainable solutions spanning distinct trade-offs (Figure 2). Evaluation on validation events of representative solutions shows that they remain competitive with scalarized single-objective results (Tables 1 and 2). Experts can therefore choose a preferred operating point after observing the alternatives.

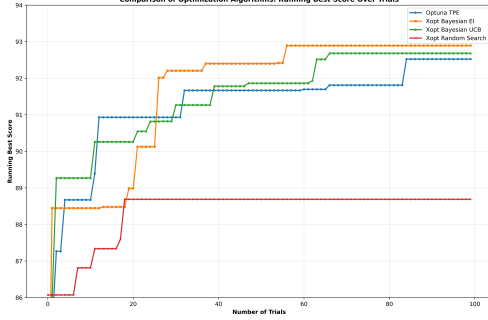


Figure 1: Running-best scalarized score on training events.

Method	N_p	S	ϵ (%)	f (%)	d (%)	t (s)
TPE	8	91.99	95.80	.250	15.25	2.551
EI	8	92.29	96.19	.252	15.51	2.735
UCB	8	92.13	96.03	.252	15.52	2.730
Random	8	89.67	96.57	.321	27.29	5.609
EI	15	92.73	96.62	.249	16.83	1.403

Table 1: Performance of the algorithms on validation events (N_p is number of optimized parameters; S is the single-objective score; ϵ , f , d , and t are efficiency, fake rate, duplicate rate, and runtime, respectively).

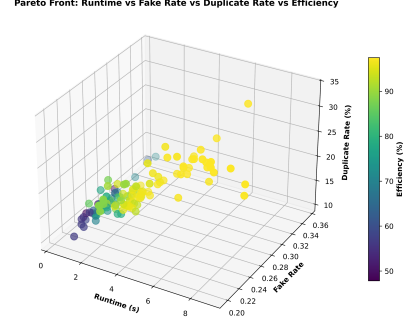


Figure 2: Non-dominated solutions on the training events in the runtime, fake rate, duplicate rate space; color represents efficiency.

Solution	A	B	C	D
ϵ (%)	96.6	96.7	96.3	96.7
f (%)	.26	.27	.26	.25
d (%)	14.7	15.8	14.6	16.8
t (s)	2.8	3.1	2.3	3.6

Table 2: Efficiency, fake rate, duplicate rate and runtime on validation events for four configurations selected from the training Pareto front.

4. Discussion and Conclusion

Together, the studies address all three gaps: Bayesian acquisition uses a fixed budget more efficiently, fifteen parameters expose a higher score, and EHVI preserves trade-offs hidden by scalarization. Scalar optimization suits known priorities; EHVI supports selection after exploration.

Results are limited by the study scale, ODD geometry, tested bounds, and 200-pileup sample; runtime is hardware dependent. The four reported EHVI configurations do not fully characterize the larger non-dominated set. Broader comparison campaigns with repeated runs are needed to constrain uncertainty across validation samples, geometries, and occupancies.

In conclusion, these results indicate that Bayesian optimization can strengthen ACTS seeding auto-tuning through efficient evaluation, larger searches that reveal better configurations, and Pareto fronts that support expert selection among physics and computing priorities.

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